Hypothesis Formulation

Learning Objectives

By the end of this session, you will be able to:

- Basics of Hypothesis Testing
- Steps of Hypothesis Testing
- Formulate H_0 and H_1
- Types of Error
- Significance Level
- Types of Statistical Tests

What is a Hypothesis and Hypothesis Testing?

- An assumption or educated guess about a population parameter.
- Tested using sample data.

- Hypothesis testing is like a detective game where you :

1. start with an assumption (a hypothesis);
2. then collect evidence (data) to decide whether
3. your assumption is likely to be true or not.

- It’s a way of making decisions or inferences about a population based on a sample of data.

Steps in Hypothesis Testing

1

1. Formulate H_0 and H_1

2

2. Choose α

3

3. Select Test

4

4. Calculate Test Statistic

5

5. Make Decision

Types of Hypotheses

✅ Null Hypothesis (H_0): This is the hypothesis that there is no effect or no difference. It's the status quo or the assumption we want to test against.

✅ Alternative Hypothesis (H_1): Opposite of null hypothesis.

When the null hypothesis is rejected the alternate hypothesis is accepted and vice versa.

Null and Alternate Hypothesis

- Null Hypothesis:
  - Hypothesis of no difference (Status Quo)
- Alternate hypothesis:
  - Mutually exclusive statement of Null hypothesis

H_0	H_a
equal (=)	not equal (≠) **or** greater than (>) **or** less than (<)
greater than or equal to (≥)	less than (<)
less than or equal to (≤)	more than (>)

Examples

Problem: Is the new drug better?

✅ H_0: New drug = Old drug

✅ H_1: New drug ≠ Old drug

Company claims average salary = ₹50,000/month.

✅ H_0: $\mu = 50,000$

✅ H_1: $\mu \neq 50,000$

Examples

Construct the null and alternative hypothesis for the following situation:

Your mobile bill per month is INR 600 per month with a standard deviation of INR 50.

This month you received a bill of INR 750.

You feel that the bill was high and you want to investigate if there is any reason for concern.

Examples

Construct the null and alternative hypothesis for the following situation:

Your mobile bill per month is INR 600 per month with a standard deviation of INR 50.

This month you received a bill of INR 750.

You feel that the bill was high and you want to investigate if there is any reason for concern.

$$H_0: \mu \leq INR600$$

$$H_1: \mu > INR600$$

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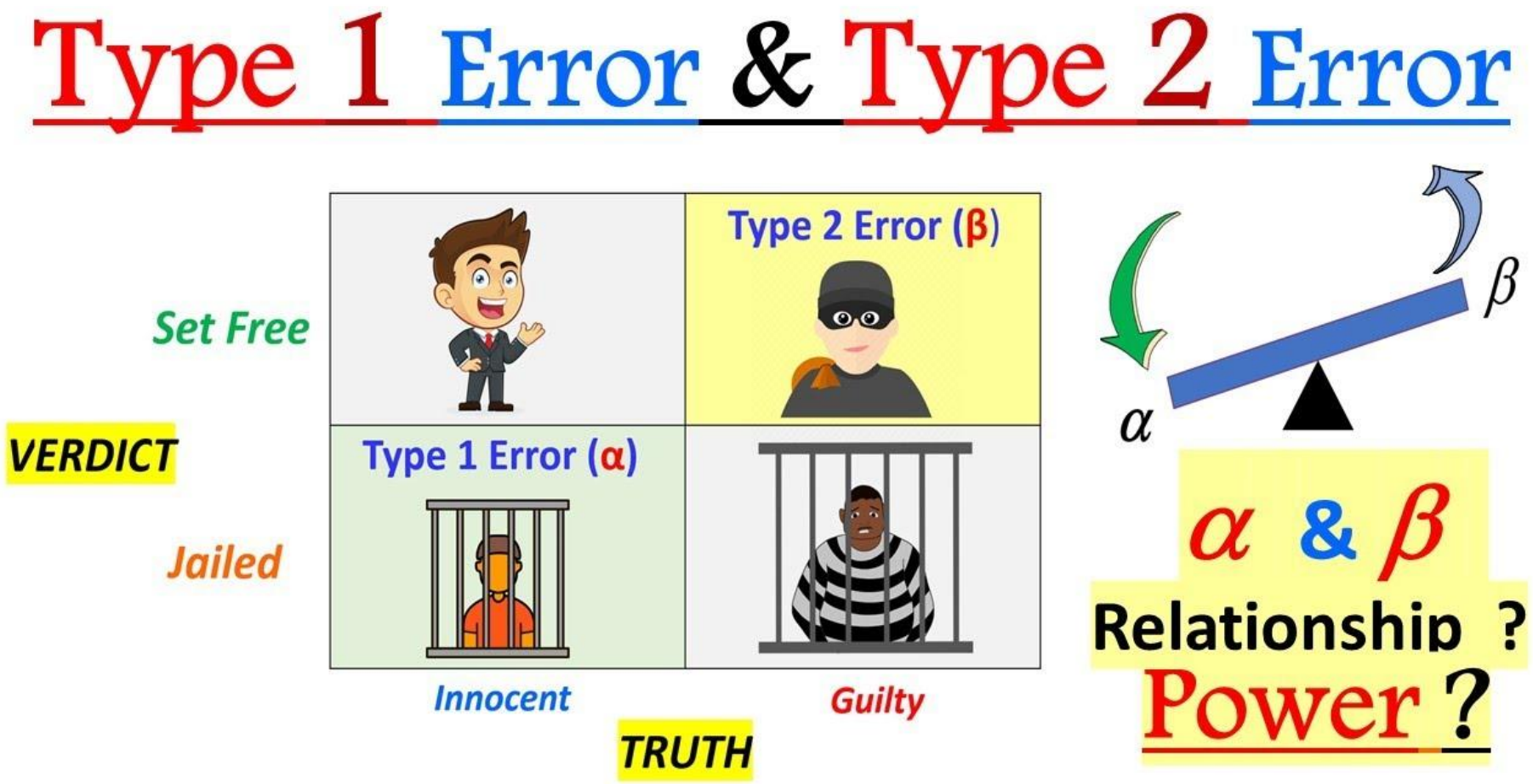
4

4. Calculate Test Statistic

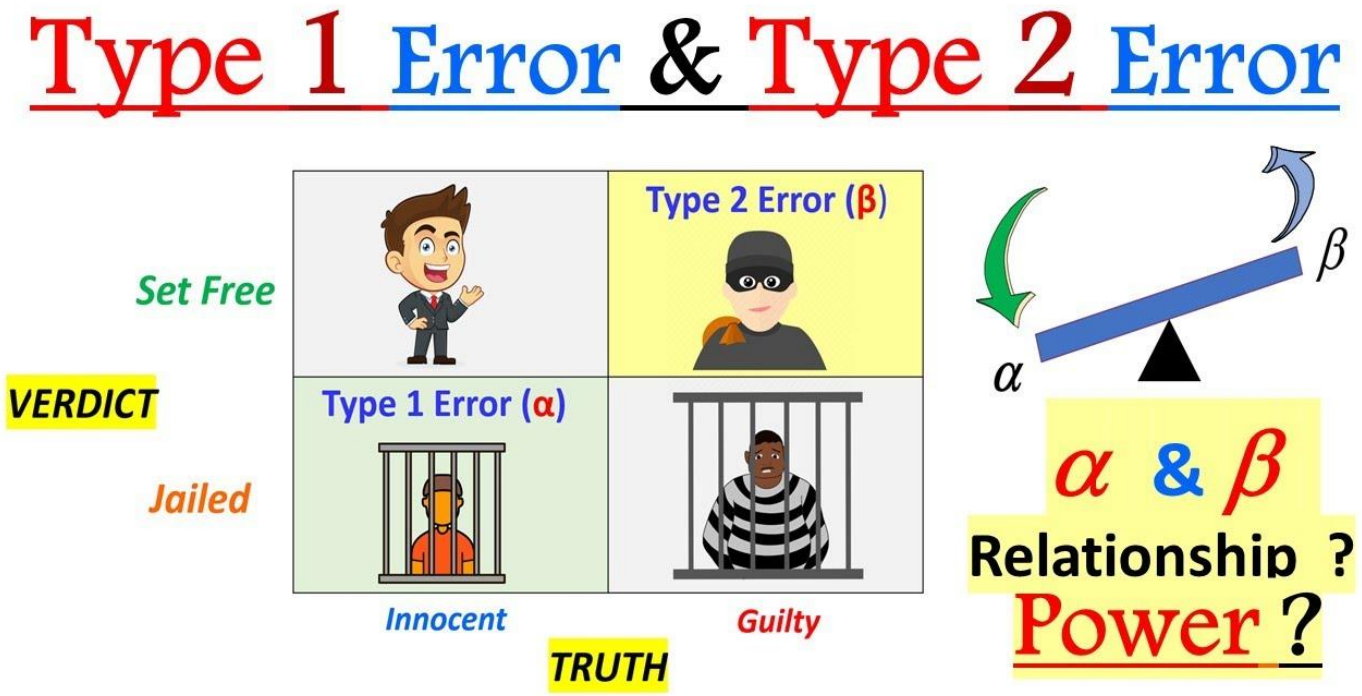
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5. Make Decision

Significance level – Type 1 and Type 2 Error



Significance level – Type 1 and Type 2 Error



Level of significance (alpha) is the probability of rejecting the null hypothesis when it is true.

$\alpha = 0.01$ or 1%

$\alpha = 0.05$ or 5%

$\alpha = 0.10$ or 10%

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Types of Tests

A **one-tail test** tests for the possibility of the relationship in one direction, either greater than or less than a certain value. **Directional (> or <)**

A **two-tail test** tests for the possibility of the relationship in both directions, whether it's significantly greater or less than a certain value. **Non-directional (≠)**

Two tail	Left tail	Right tail
H0: Mean=10 Ha: Mean≠10	H0: Mean ≥ 10 Ha: Mean < 10	H0: Mean ≤ 10 Ha: Mean >10

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Test Statistic

A **test statistic** is a standardized value derived from sample data during a hypothesis test.

It helps determine whether to reject the null hypothesis.

Different tests (t-test, chi-square, etc.) have different formulas for calculating their respective test statistics.

$$\textbf{Test statistics} = \frac{\textit{Sample statistic} - \textit{Population Parameter}}{\textit{Standard Error}}$$

Test Statistic

We want to test that the population mean is different than 10

We want to test that the population mean is less than 10

We want to test that the population mean is greater than 10

Two tail	Left tail	Right tail
H0: Mean=10 Ha: Mean≠10	H0: Mean ≥ 10 Ha: Mean < 10	H0: Mean ≤ 10 Ha: Mean >10

Commonly used Critical z values

Significance Level (α)	Two-Tailed Test	One-Tailed Test (Left)	One-Tailed Test (Right)
0.01	±2.58	-2.326	+2.326
0.05	±1.96	**-1.645**	**+1.645**
0.10	**±1.645**	-1.282	+1.282

MCQ

1. What is H_0 usually about?

A) No change

B) New discovery

C) 5% significance

2. Which error occurs when H_0 is wrongly rejected?

A) Type I

B) Type II

C) Type III

MCQ

1. What is H_0 usually about?

A) No change ✅

B) New discovery

C) 5% significance

2. Which error occurs when H_0 is wrongly rejected?

A) Type I ✅

B) Type II

C) Type III

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Session Summary

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- Types of Error
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Reference links

- Introduction to Hypothesis Testing
- Frequently asked hypothesis testing interview questions and answers – part 1
- Frequently asked hypothesis testing interview questions and answers – part 2
- Frequently asked hypothesis testing interview questions and answers – part 3